

Problem Statement #37

Group Travel Expense Simplification Engine

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Deliverables:

1. Requirements Table (5 FR + 2 NFR)
2. UML Use-Case Diagram
3. Use-Case Flow Specification (UC-003)

1. Functional Requirements (FR)

ID	Description	Priority	Acceptance Criteria
FR-001	The system shall allow users to record group expenses with uneven split ratios (percentage, exact amount, or equal share) across multiple members in a single transaction.	High	Pass: A 4-member group can split a \$200 expense as 50%, 25%, 15%, 10%; system stores each member's liability accurately. Fail: Split total does not equal expense amount or rounding errors exceed \$0.01.
FR-002	The system shall automatically convert foreign currency expenses to a trip's base currency using daily exchange rates fetched from a reliable external API (e.g., Open Exchange Rates, ECB).	High	Pass: A €150 expense recorded in a USD-based trip converts correctly at the day's closing rate; converted amount stored with 4-decimal precision. Fail: Rate is older than 24 hours or conversion error exceeds 0.5%.
FR-003	The system shall compute the optimal debt settlement graph using a min-flow algorithm to minimize total peer-to-peer payment transactions required to zero out all group balances.	High	Pass: For net positions [+30, -10, -20], system outputs 2 transactions; sum of debts equals sum of credits; zero residual balance. Fail: Redundant transactions present or net group balance != 0.
FR-004	The system shall provide a real-time group balance dashboard displaying each member's net position (owed/owing) in the trip's base currency, updated immediately after every new expense or settlement.	Medium	Pass: Dashboard reflects updated balances within 2 seconds of transaction commit; color-coded indicators. Fail: Stale data persists beyond 5 seconds or incorrect net position displayed.
FR-005	The system shall support trip lifecycle management allowing a Trip Admin to create a trip, invite members via email/link, set a base currency, and archive the trip upon completion.	Medium	Pass: Admin creates trip, invites 3 members, sets EUR as base; invited members receive notification and can join. Archived trips are read-only. Fail: Non-admin members modify trip settings or archived trips remain editable.

2. Non-Functional Requirements (NFR)

ID	Type	Description	Priority	Acceptance Criteria
NFR-001	Performance & Scalability	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 recorded expenses under normal server load.	High	Pass: Load testing shows 95th percentile latency < 100 ms; throughput >= 100 req/sec. Fail: Latency exceeds 150 ms or timeout errors under peak simulation.
NFR-002	Security & Data Integrity	All expense data and user PII must be protected using AES-256 encryption at rest, TLS 1.3 in transit, and comply with GDPR data retention policies.	High	Pass: Security audit confirms encrypted DB fields, valid SSL, no plaintext PII in logs; GDPR right-to-erasure completes within 30 days. Fail: Unencrypted data in backups or unauthorized API access leaks sensitive data.

3. UML Use-Case Diagram

Actors:

- Group Traveler: Participates in trips, records expenses, views balances, and initiates settlements.
Trip Admin: Creates trips, invites members, manages settings, and has elevated privileges for expense oversight.

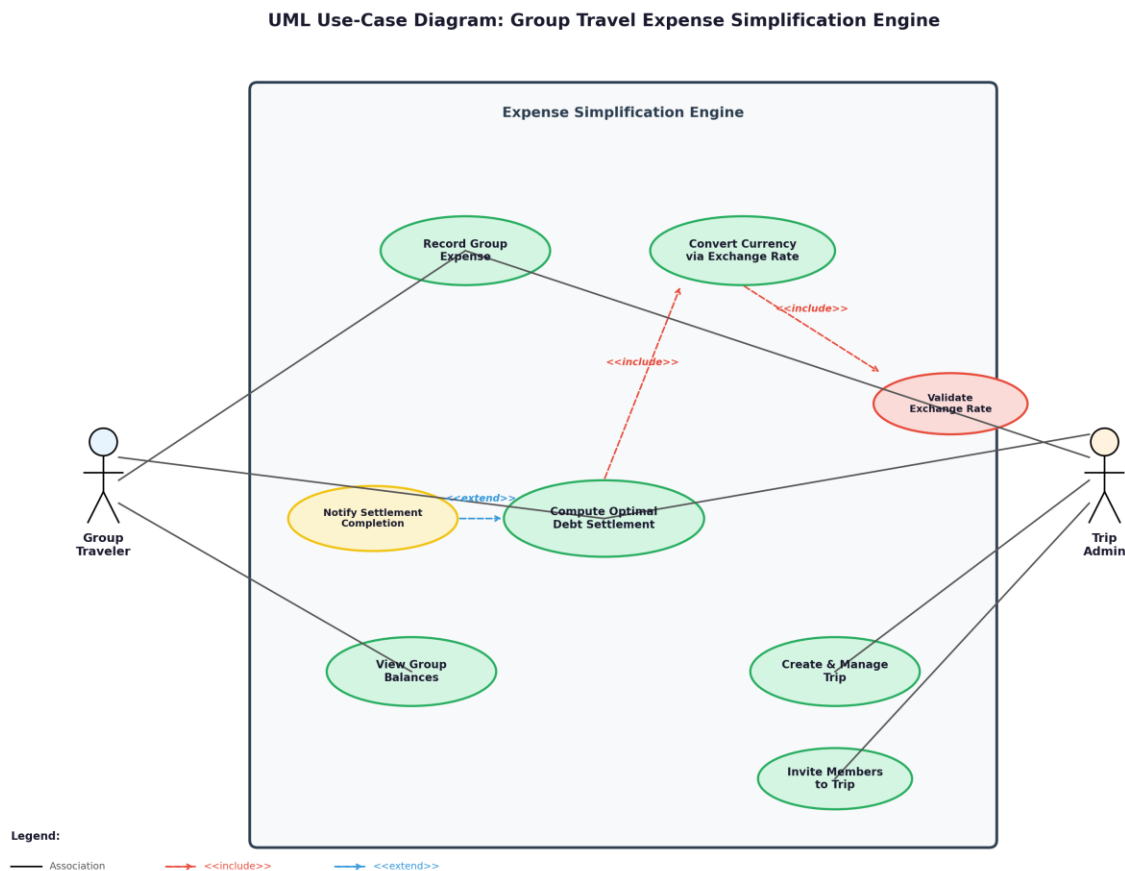
Primary Use Cases:

- Record Group Expense
Convert Currency via Exchange Rate
Compute Optimal Debt Settlement
View Group Balances
Create & Manage Trip
Invite Members to Trip

Relationships:

<<include>>: Convert Currency via Exchange Rate includes Validate Exchange Rate (mandatory freshness check).
Compute Optimal Debt Settlement includes Convert Currency via Exchange Rate (normalization required before settlement).

<<extend>>: Notify Settlement Completion extends Compute Optimal Debt Settlement (triggered only if user opted in for notifications).



4. Use-Case Flow Specification

UC-003: Compute Optimal Debt Settlement

Use-Case ID:	UC-003
Use-Case Name:	Compute Optimal Debt Settlement
Actor(s):	Group Traveler, Trip Admin
Description:	The system calculates the minimum set of peer-to-peer transactions required to settle all outstanding debts within a trip group, converting multi-currency liabilities into the trip's base currency before running the optimization algorithm.

Preconditions:

2. User is authenticated and belongs to an active trip.
3. Trip contains at least one recorded expense.
4. All foreign-currency expenses have associated exchange rates (cached or freshly fetched).
5. User has permission to view/settle balances.

Postconditions:

6. A settlement graph is generated and persisted.
7. Group net balance equals zero.
8. Each member can view their specific pay/receive instructions.
9. Transaction count is minimized (provably optimal for given inputs).

Main Success Scenario (Basic Flow)

Step	Actor Action	System Response
1	Group Traveler navigates to the trip's "Settle Up" tab and clicks "Compute Settlement".	System validates user membership and trip status.
2	System: Retrieves all unarchived expenses for the trip.	System retrieves all unarchived expenses for the trip.
3	System: For each foreign-currency expense, invokes Convert Currency via Exchange Rate (UC-002).	For each foreign-currency expense, system includes Convert Currency via Exchange Rate (UC-002) to normalize values into base currency.
4	System: Aggregates per-member net positions: sum(amounts paid) minus sum(share of liabilities).	System aggregates per-member net positions: sum(amounts paid) minus sum(share of liabilities).
5	System: Executes the min-flow debt simplification algorithm on the net position vector.	System executes the min-flow debt simplification algorithm on the net position vector.
6	System: Generates the settlement graph: directed transactions (payer → payee, amount).	System generates the settlement graph: list of directed transactions (payer -> payee, amount).
7	System: Persists the settlement plan and updates trip status to "Pending Confirmation".	System persists the settlement plan; updates trip status to "Pending Confirmation".
8	System: Displays settlement graph with payer/payee names, amounts, and optional payment links.	System displays settlement graph with payer/payee names, amounts, and optional payment links.
9	System: If notifications are enabled, extends with Notify Settlement Completion.	If notifications enabled, system extends with Notify Settlement Completion.
10	Group Traveler reviews the proposed transactions and clicks "Confirm Settlement".	System marks settlement "Confirmed"; archives underlying expenses.

Alternate Flow - Stale Exchange Rates

Step	Actor Action	System Response
3a	System: Detects that the exchange rate is older than 24 hours or missing.	During currency conversion, system detects exchange rate older than 24 hours or missing.
3a.1	System: Invokes Validate Exchange Rate; validation fails.	System includes Validate Exchange Rate; validation fails.
3a.2	System: Fetches fresh rates from the external API and retries conversion.	System fetches fresh rates from external API; retries conversion.
3a.3	System: If the API is unreachable after 3 retries, prompts the user to use the last known rate or try later.	If API unreachable after 3 retries, prompts: "Exchange rates unavailable. Use last known rate or try later?"
3a.4	Group Traveler selects "Use Last Known Rate" or "Cancel".	If "Use Last Known": proceed to Step 4 with cached rate flagged. If "Cancel": abort; return to dashboard.

Alternate Flow - Group Already Balanced

Step	Actor Action	System Response
5a	System: Determines that all net positions are zero.	After aggregation, all net positions are zero (group already settled).
5a.1	System: Displays: "All balances are already settled. No transactions needed."	System displays: "All balances are already settled. No transactions needed."
5a.2	Group Traveler acknowledges message and returns to dashboard.	Use case terminates successfully; no settlement graph generated.

Exception Flow - Algorithm Timeout

Step	Actor Action	System Response
5b	System: Detects that the min-flow algorithm exceeds the NFR-001 threshold.	For unusually large group (>50 members or >500 expenses), min-flow algorithm exceeds NFR-001 threshold (100 ms).
5b.1	System: Falls back to greedy approximation and displays the large-group warning.	System falls back to greedy approximation and displays warning: "Large group detected. Using optimized approximation."
5b.2	Group Traveler reviews approximate settlement graph and decides to proceed or cancel.	Proceeds to Step 6 with approximate result, or cancels and returns to dashboard.